

Project Setup and Run Instructions

This document provides detailed instructions on how to setup and run the project. Please follow the steps carefully to ensure the application runs correctly.

Prerequisites

Before starting, make sure your system meets the following requirements:

1. **MYSQL:** Install MySQL version 8.0 or higher on your computer. You can download it from the [MySQL official website](#). Make sure that the root user has password "root". Otherwise, the app cannot connect to the database.
2. **Docker:** Install Docker and Docker Compose. You can find installation instructions at the [Docker official website](#)
3. **Visual Studio Code (VS Code):** Install VS Code to open and manage the project files. You can download it from [Visual Studio Code official website](#)
4. **Node.js:** Make sure Node.js and npm are installed on your system. You can download it from [Node.js official website](#)

Steps to setup the project

1. After extracting the folder, you can now open the project folder in VS Code.
2. Now you need to install the dependencies (they were removed to ensure smaller file size of the project):
 - a. Open the terminal in VS Code by selecting "Terminal" -> "New Terminal" from the top menu
 - b. Run the following command to install all required dependencies:
3. Now you need to start Docker. First, open the application or make sure it is running on your system. After that you have to follow these steps:
 - a. Navigate to the Docker folder inside the root project directory. Use the terminal to move to this folder:

```
npm run install-all
```

- c. This process may take a few minutes. Wait until all installations are complete. You may see messages about some packages being outdated or deprecated, or you can get vulnerabilities warning but just ignore them.

```
cd docker
```

- b. Run the following command to start the Docker container using the docker-compose.yml file:

```
docker-compose up
```

- c. This will start necessary containers for the application. When you want to shut it down, just press CTRL + C and type “y” when prompted.
- 4. The last step is to run the development server. You need to open a new terminal for this because the Docker uses the previously opened terminal. Now you need to follow these steps:
 - a. Start the development server by running:

```
npm run dev
```

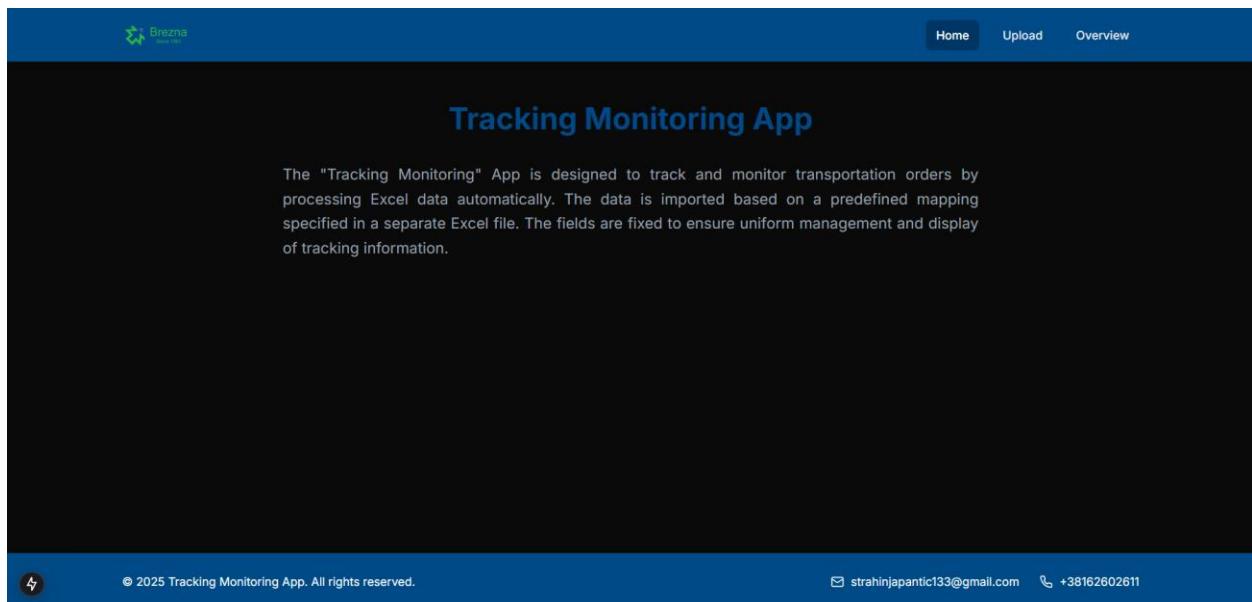
- b. After all the services are started, you can access the Web Application by opening your browser and navigating to:

```
http://localhost:3000
```

Explanation of the setup: I have concurrent package inside my root package.json file, which allows the user to easily install all dependencies at once for all 4 services just by running 1 command. It also simplifies running the development server by running all the commands for starting the services with just 1 command. Docker container can also be started via desktop application so that it does not need a dedicated terminal.

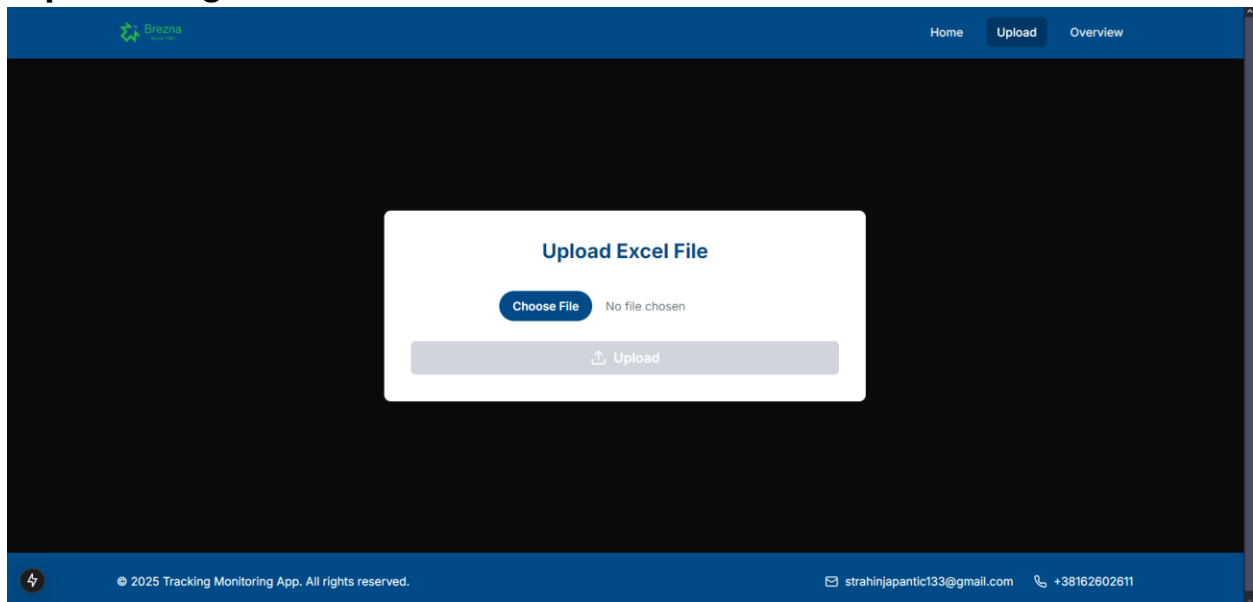
Usage Instructions

Once you open the web application via link: <http://localhost:3000> you should see a home page, which looks like this:

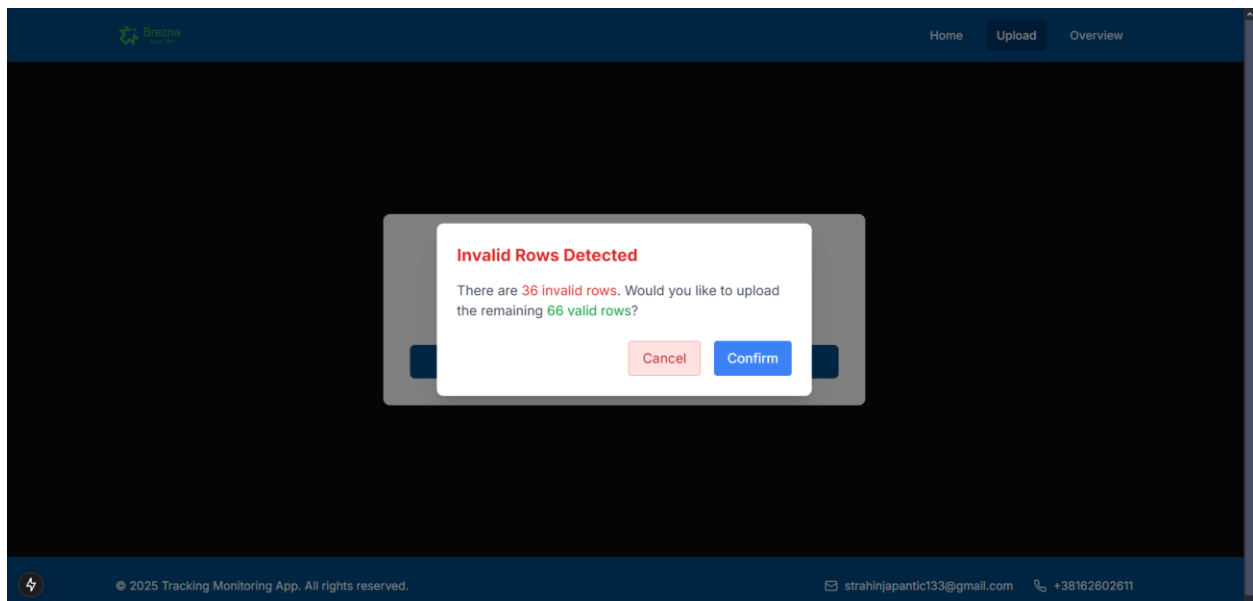


You can see in the navbar that there are 3 links. First is for the Home page, second is for the Upload page and third is for the Overview page.

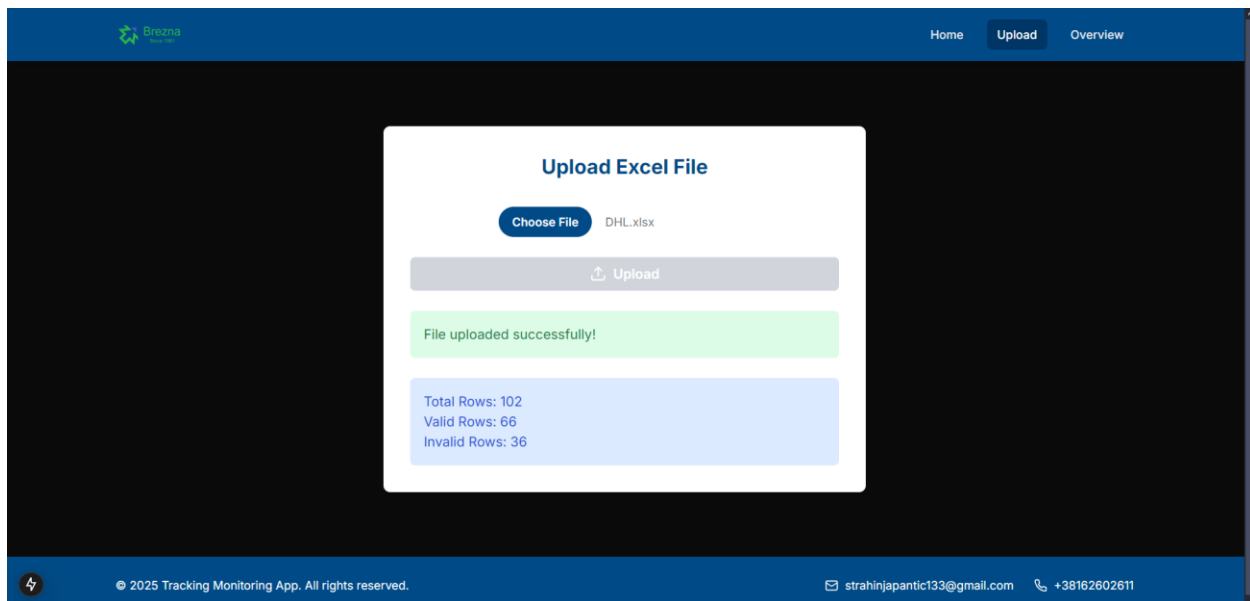
Upload Page



Once you open the upload page, there is a “Choose File” button, which will let you upload only .xlsx and .xls files. If you upload DHL.xlsx file and click the “Upload” button, then you should get a confirmation dialog, which looks like this:



If you click on the button “Cancel”, then nothing will happen and you would get a message saying “**Upload cancelled by user.**”. However, if you click on the “Confirm” Button, you will get a screen like this:

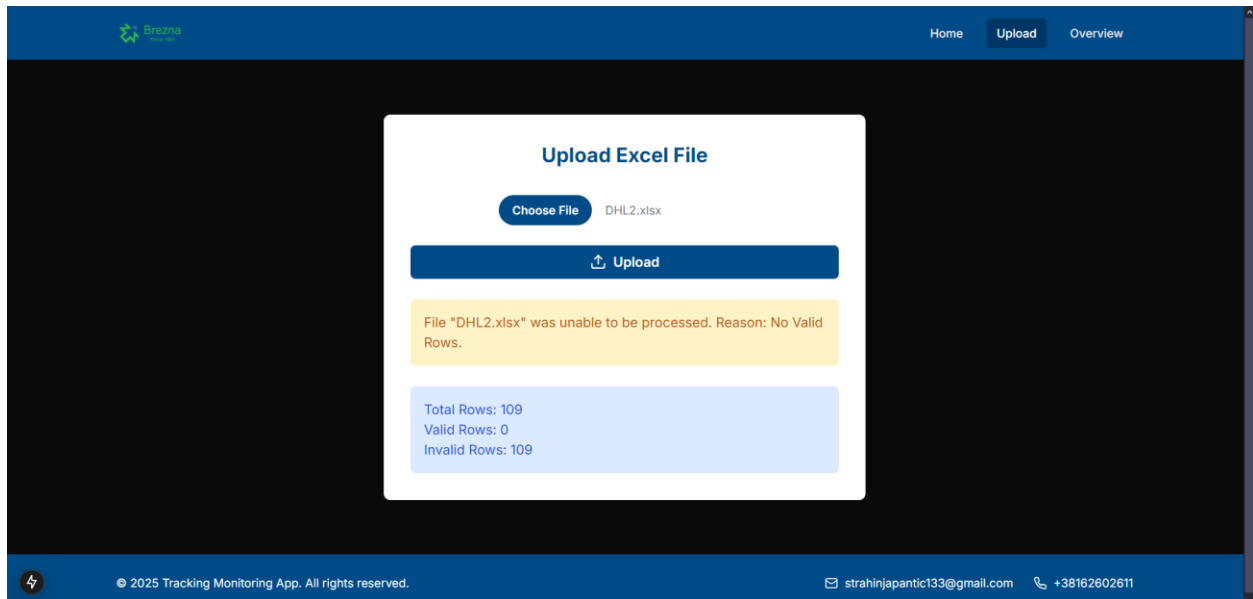


This means that the file has been successfully uploaded, processed and the valid rows were inserted in the database. Because you clicked “Confirm” button this time, you will also get a log file for this upload, which is located in this directory: `services/file-processing-service/logs`.

This similar behavior will occur when you try to upload `hellmann.xlsx` and `hellmann2.xlsx` files, but it will just take a little bit more time. If you get a timeout error when uploading `hellmann` files, then you can change the timeout timer to a greater value inside this file: `services/api-gateway/src/app.module.ts`. Timeout error will be seen in the terminal output inside VS Code. If you see Unknown error in the web app, then check the console.

```
services > api-gateway > src > TS app.module.ts > AppModule
1  /**
2   * app.module.ts
3   * Root module of the API Gateway
4   * Configures dependencies and defines the application structure
5   */
6  import { Module } from '@nestjs/common';
7  import { ConfigModule } from '@nestjs/config';
8  import { HttpModule } from '@nestjs/axios';
9  import { HealthController } from './controllers/health.controller';
10 import { ProxyController } from './controllers/proxy.controller';
11 import { ProxyService } from './services/proxy/proxy.service';
12
13 @Module({
14   imports: [
15     // Configure environment variables globally
16     ConfigModule.forRoot({
17       isGlobal: true, // Make config available throughout the app
18       cache: true, // Cache config values for better performance
19     }),
20     // Configure HTTP client for making requests to other services
21     HttpModule.register({
22       timeout: 60000, // 60 second timeout
23       maxRedirects: 5, // Allow up to 5 redirects
24       maxContentLength: Infinity, // No limit on response size
25       validateStatus: (status) => status < 500, // consider only 500+ as errors
26     }),
27   ],
28   controllers: [HealthController, ProxyController],
29   providers: [ProxyService],
30 })
31 export class AppModule {}
```

If you upload DHL2.xlsx file you will get a screen like the image below. That is normal because there are no valid rows found in that file which means that there is nothing to upload.



When you open the Overview page, you will see a table with 50 rows and pagination controls. You can click on any of the rows inside that table which will open the details page about that record. The detail page will display every field from the database for that record. If that field is null in the database, then it will just display “—” sign. Above that detailed information is the link “Back to Overview” which can bring you back to the Overview page. All data fields with dates and time is displayed in the same format: dd/mm/yyyy hh:mm:ss.

If you want to check the logs after uploading, you can navigate to folder `level_up_project/services/file-processing-service/logs` and inside it you will see the logs are created with detailed information about the upload.

Warnings

If you already have a MySQL installed on your computer, make sure your root user has a password “root” or the application will not be able to connect to the database.

Make sure that ports: 3000, 3001, 3002, 3003, 3306 and 3307 are free because those are the ports used by this application.